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Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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❖ CODE :

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++) frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

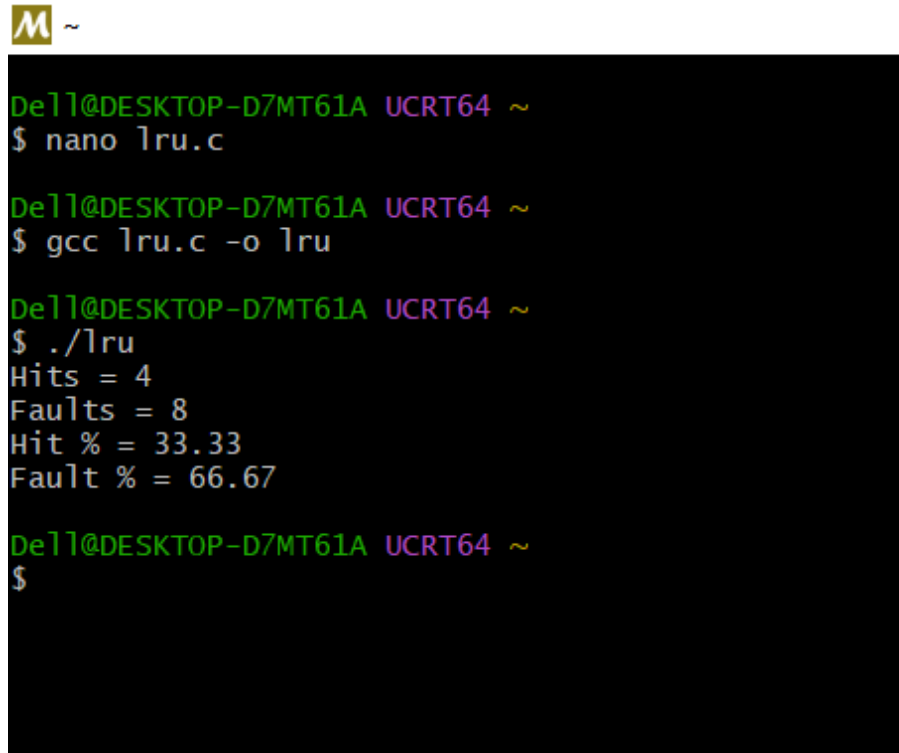
            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0]; pos = 0;
                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
                }
                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}
```

AG Help AO Write Out AF Where Is AK Cut AT Execute AC Loc
AX Exit AR Read File AN Replace AU Paste AJ Justify AL Go

❖ OUTPUT :



```
M ~  
De11@DESKTOP-D7MT61A UCRT64 ~  
$ nano lru.c  
De11@DESKTOP-D7MT61A UCRT64 ~  
$ gcc lru.c -o lru  
De11@DESKTOP-D7MT61A UCRT64 ~  
$ ./lru  
Hits = 4  
Faults = 8  
Hit % = 33.33  
Fault % = 66.67  
De11@DESKTOP-D7MT61A UCRT64 ~  
$
```